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A magnetic anomaly study offshore the Canary Archipelago

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Abstract

A scalar magnetic anomaly map of the regions offshore the Canary Archipelago is presented here. This map is based on measurements taken inside the Exclusive Economic Zone Project framework. This paper contains a description of the data set, whose accuracy and internal consistency are analysed. The magnetic anomalies are described and the main structural trends are highlighted. This analysis has served to detect two possible fractures with a NW–SE component: one separates Fuerteventura from Gran Canaria Island, and the other apparently runs through Fuerteventura. The latter finding agrees with a 1.7 km depth offset, between its northern and southern halves, detected by the algorithm based on the Euler Deconvolution. A similar approach has been performed, estimating the depth to the top of the most significant anomalies. This result shows that the most noteworthy magnetic anomaly source of the archipelago, which lies between Tenerife and Gran Canaria Island, seems to be located at an average depth of 4 km below sealevel. A spectral analysis was performed to estimate the depth extent of the deepest anomalies. It argues the presence of sub-crustal magnetic sources (underplating) in the archipelago. A standard Euler Deconvolution analysis was executed to analyse the spatial distribution of these mantle-like sources. Our results seem to support the existence of magmatic underplating under Gran Canaria, Tenerife, and Fuerteventura, and suggest this possibility for La Palma.

Introduction

This paper focuses on the processing and inversion of magnetic anomaly data from the Canary Archipelago, inside the Exclusive Economic Zone framework. We have produced a scalar magnetic anomaly map, and, by applying some operators (e.g., analytical signal and Euler Deconvolution), we have highlighted structural features and trends around this area, and inferred the depth and location of the causative bodies. These results improve the resolution and expand on earlier work in the area. Additionally, the magnetic data can now be correlated with other data with the same level of accuracy and resolution.

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Magnetic geological setting

The Canary Archipelago is located on the border of the northwestern Africa passive continental margin. Its geographical location indicates that the oceanic crust on which these islands were built is Early Jurassic in age (150–170 Ma). To improve our understanding of this zone we have compiled magnetic data from the Geophysical Data System (GEODAS) (Metzger and Campagnoli, 2003) and merged them into a single 0.2° × 0.2° grid covering the following geographical frame: 35°N–25°N and 5°W–25°W (Figure 1).

Figure 1 shows a magnetic anomaly pattern which is clearly unbalanced and is dominated by a spectacular feature, M0–M4 anomalies, previously known as the J anomaly (Hayes and Rabinowitz, 1975). Its shape delineates a N30°E trend, showing a quite remarkable continuity northward. Moving eastward, other M-series spreading anomalies (black solid line in Figure 1) stand out, e.g., M16, M21 and M25, the oldest alignment consistently present in the Eastern Atlantic (Vogt, 1986; Roest et al., 1992a).

The M0–M4 series lose definition and amplitude at the Canary Lineament (near 30°N paral-

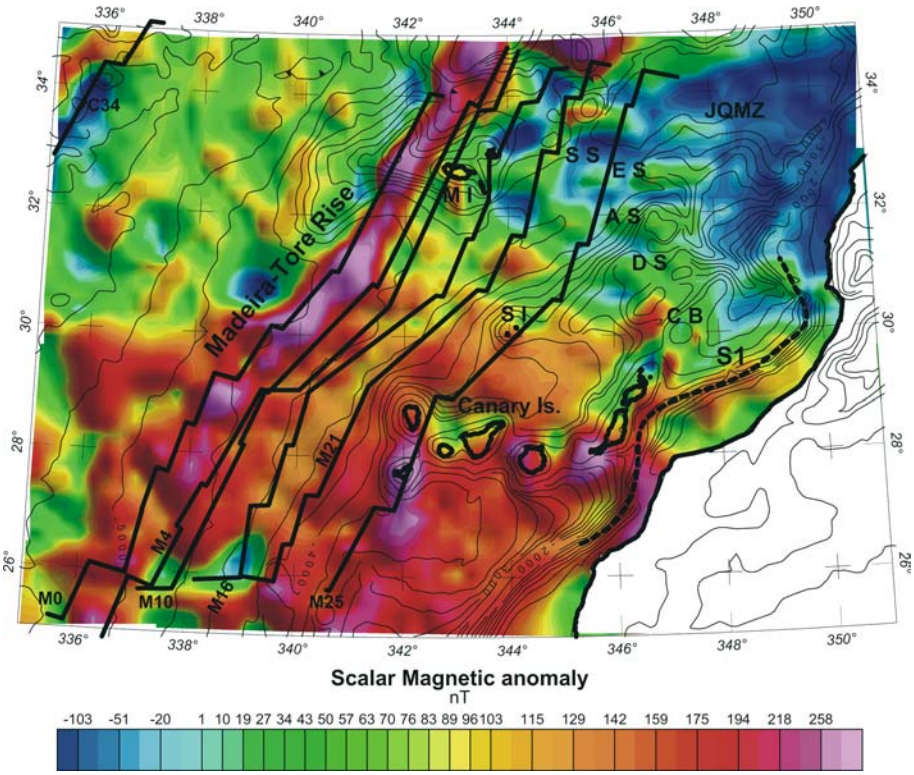

Figure 1. Shaded relief map of scalar field magnetic anomalies, illuminated from the northwest. Figure shows principal features: volcanic islands, linear seafloor spreading magnetic anomalies after Verhoef et al. (1991), in solid lines bathymetry, contour interval: 200 m. CB: Conception Bank; DS: Dacia Seamount; SI: Selvagens islands; MI: Madeira islands; SS: Seine Seamount; ES: Essaouira Seamount; AS: Agadir Seamount, JQMZ: Jurassic Quiet Magnetic Zone. Map projection is Lambert Conic Conformal (scale 1:10.000.000).

lel). A magnetic quiet zone is observed landward of the M-sequence throughout the whole area (Jurassic Magnetic Quiet Zone (JMQZ), 150–170 Ma). Areas of disturbances almost always correspond to the presence of volcanic islands or seamounts, as was inferred from the correlation with bathymetry or with prominent basement peaks (Hayes and Rabinowitz, 1975). By including the Madeira and Canary Archipelagos, Figure 1's geographical frame contains the highest density of islands/seamounts on the African continental margin (Ye et al., 1999).

Eastward of anomaly M25, inside the JMQZ, no seafloor spreading anomalies are visible. Only an elongated anomaly, just off the African margin, can be detected easily. This anomaly, which has a remarkably similar geometry to that of anomaly M25, could be associated with the so-called slope anomaly S1. This anomaly could mark the approximate landward edge of the oceanic crust (Roest et al., 1992a; Roeser et al., 2002).

Previous magnetic investigations

The Canary archipelago has been the subject of many geophysical investigations, but not much magnetic study. In 1993, the Spanish National Geographical Institute (I.G.N.) carried out an aeromagnetic survey composed of seven independent grids flown at different heights and in a north-southward orientation. For further technical details see Socias and Mezcua (1996). As a final result, an aeromagnetic map at 3200 m was published in 1995 (Instituto Geográfico Nacional, 1995).

In April 1995, a marine magnetic survey was carried out in the Tenerife offshore area by the Spanish Oceanographic Institute (I.E.O.). This geophysical campaign, called "Teide-95", included not only magnetic, but also bathymetric and gravity data acquisition. Its main aims involved the mapping of offshore area morphology, trying to determine the volume of pyroclastic rocks and extend the survey as close to the coast as possible.

Ultimately, the goal was to define the relationships between the offshore slope morphology and some onshore collapsed valleys in Tenerife (Teide Group, 1997).

Araña et al. (2000) established a structural model for Tenerife, combining mainly magnetic and gravity data, as well as other volcanological and similar geophysical data. Blanco-Montenegro et al. (2003) performed a similar study focused on magnetic data. They propose a crustal structure model beneath Gran Canaria island and suggested, from spectral analysis, the presence of rocks at mantle-like depths (from the Moho to about 23 km) that could behave as magnetic sources. They discussed the possible relation with magmatic underplating proposed by previous studies (Ye et al., 1999; Dañobeitia and Canales, 2000).

Verhoef et al. (1991) published a magnetic study performed after compilation of magnetic data collected off West Africa over a period of nearly 30 years (1961–1988). It allowed the accurate identification of sea floor spreading anomalies, showing offsets and disruptions in the anomalies caused by fracture zones. It not only focused on the Canary islands but covered a wider area, limited to the north by the Azores Gibraltar Fracture Zone (AGFZ, 37°N) and to the south by the Kane Fracture Zone (KFZ, 21°N). This dataset, in combination with data from the western Atlantic, was used in later work to decipher the African and conjugate margin spreading histories (Roest et al., 1992a).

Since 1998, the Spanish Exclusive Economic Zone Project (ZEEE) has studied the Canary Islands area. This paper presents a magnetic anomaly map derived from the compilation of scalar magnetic measurements. They were obtained during three one-month marine surveys during 1998, 1999 and 2000 carried out in the Canary Islands area, aboard the Spanish Research Vessel "Hespérides". Two additional campaigns, undertaken in October 2001 and 2002, used the Spanish Oceanographic Ship "Vizconde de Eza".

Data compilation and methods

Instrumentation

In order to measure the geomagnetic field at sea, three marine magnetometers were used: Geomet-

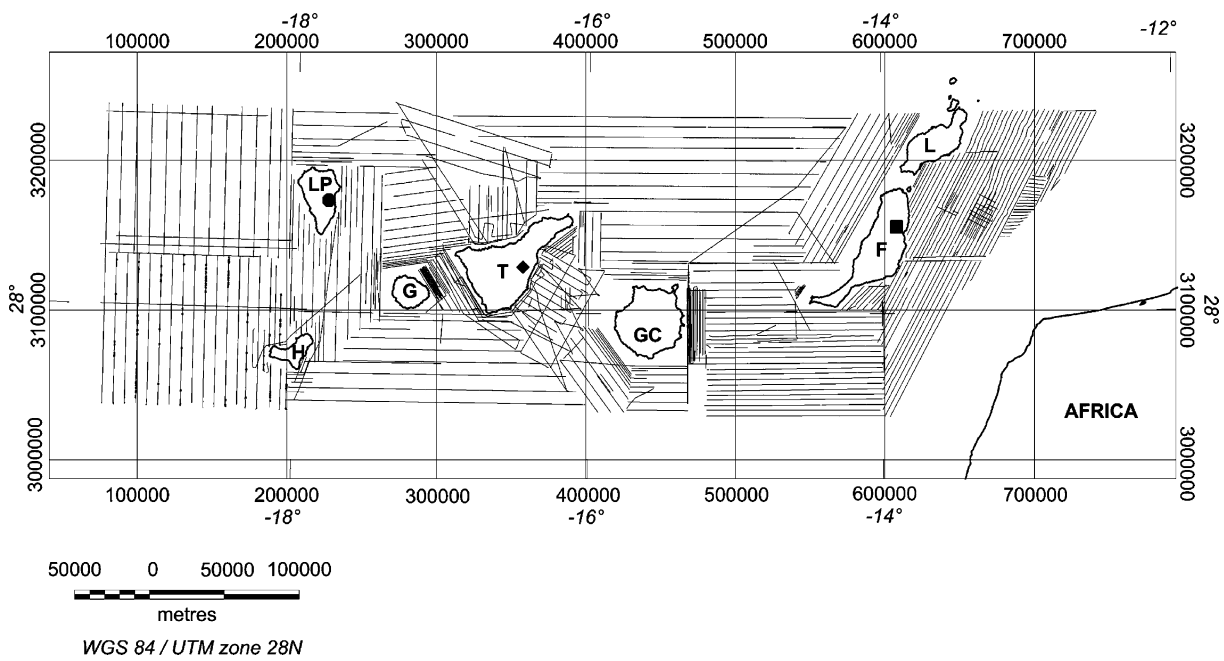

Figure 2. ZEEE track line in the region of the Canary islands (1998–2002). Reference station sites are included, using square and circle symbols respectively for Fuerteventura and La Palma. Güimar observatory (Tenerife) is marked with a diamond symbol. H: Hierro; LP: La Palma; G: Gomera; T: Tenerife; GC: Gran Canaria; F: Fuerteventura; L: Lanzarote.

rics *G-801* Proton Precession Magnetometer, (ROA), Geomag *SMM II* Overhauser effect magnetometer (provided by the University of Cadiz), and a Geometrics *G-876* type, installed on the B.I.O. "Hespérides".

To ascertain the external geomagnetic field component, three reference station sites were used on land (Figure 2). Güimar Observatory (I.G.N.) on Tenerife island provided data during the whole survey period. Additionally, a portable reference station was installed and maintained by ROA staff on La Palma and Fuerteventura. Their periods of activity were adapted to the surveying. The permanent presence of Güimar Observatory, considering its geometric emplacement, allowed us to infer the external field error budget contribution. This difference has been checked and proved not to be greater than 5 nT (in root mean square sense) even during magnetic storm periods. This unexpected result could be justified by the survey shape, which mainly evolved in a Westward progression.

Error estimates and data compilation

In order to evaluate the precision of the surveys and the internal consistency of the whole data set, a crossover analysis was performed. The results, from 639 crossings, show a standard deviation of 12.9 nT and a mean value of less than 0.5 nT. Thus, we can infer that the results have not been influenced by any systematic effect.

Results are gathered in the following histogram (Figure 3). Residuals could be due to several factors:

1. An incomplete cancellation of the diurnal variation.
2. An indirect effect produced by a possible lack of precision in ship's positioning. Considering the magnetic field local gradient, incorrect positioning could generate differences in crossover readings near the coast, where the spatial gradient reaches maximum values.
3. Internal field contribution.
4. Error due to the secular variation models used.
5. Instrumental errors.

The data has been subjected to classical corrections, dating both in time and position using precise satellite navigation (DGPS). Its precision (better than 20 m) induces an expected error contribution for the navigation uncertainty not greater than 1 nT. All tracks were corrected using the Definitive Geomagnetic Reference Field

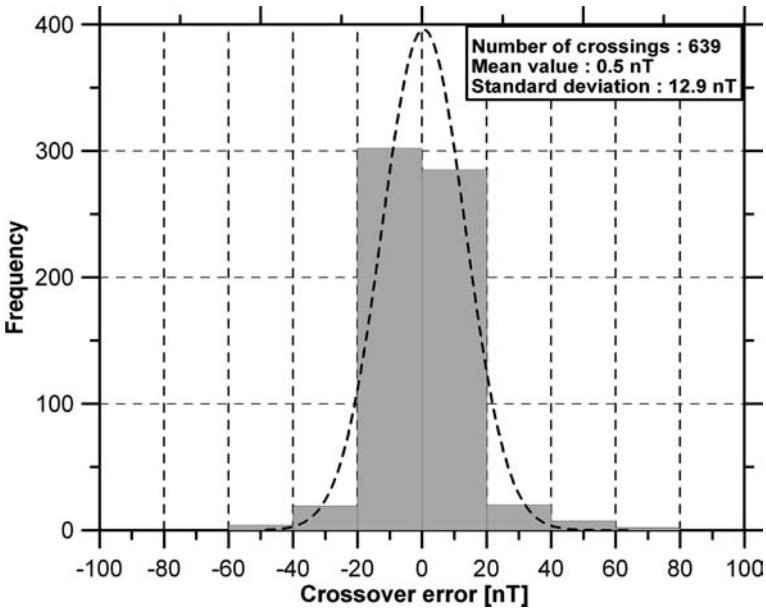

Figure 3. Histogram of crossover errors.

model DGRF-95 for the period 1998–2000 and the IGRF-2000 for 2000–2002. All measurements were referred to 2000.0 epoch.

In order to take into account a possible error contribution due to discrepancies caused by the use of two different regular field models (DGRF-95 and IGRF-2000), we have compared two grids built with the different regular field models used, dated at 2000.0, using the Canary Archipelago's geographical frame. The results conclude that this contribution is smaller than 1 nT and is therefore not significant.

Survey coverage was not uniform. It consisted of irregularly spaced and irregularly oriented ship tracks (Figure 2) with an across-track distance which ranging from 10 km to 500 m, with an approximate 30 m along-track resolution conditioned by local water depth. To convert this into observations over a regular grid, we have applied a Kriging algorithm, which suits our needs quite well when we merge different non-parallel line data. Finally, a 5 km × 5 km grid cell size was used.

To avoid possible edge effects we have digitised the 3200 m aeromagnetic map (Instituto Geográfico Nacional, 1995) shoreline contour anomalies and downward continued these new magnetic anomaly values to sea level. There is no question about instabilities due to this downward continuation because no magnetic sources were cut (Blakely, 1995).

Particular attention must be paid to the aeromagnetic anomaly data used. The aeromagnetic anomaly grid was originally obtained using a IGRF-85 model, and referred to the epoch 1993.8 (Instituto Geográfico Nacional, 1995). In order to avoid the application of the IGRF-85 model on an epoch outside its validation period (1985–1990), we have removed this analytical model, finally extracting the regular field by use of the DGRF-90 evaluated in the flight epoch (1993.8).

As required by a spectrum analysis explained below, we need to fill in those gaps that correspond to the different islands' land coverage. We have digitised seven 5 km × 5 km grids (one grid per island) from the 3200 m aeromagnetic map (Instituto Geográfico Nacional, 1995) and proceeded in the same way as described above to avoid the IGRF-85 model and extract data with the DGRF-90 model. We have upwardly continued our shipborne data until 3200 m and merged them with the aeromagnetic data, obtaining a new grid without any geographical gap.

In order to be certain that no magnetic sources actually exist in the region of continuation, we have upwardly continued our new 3200 m grid up to the highest summit height (the Teide peak: 3800 m, in Tenerife). Not doing so could have introduced some type of instability (Blakely, 1995).

Magnetic anomaly map

Description of the map

As described in the previous paragraph, we were able to set up a 3800 m grid for the whole archipelago, but we decided not to use it, except for two specific instances in order not to lose spatial resolution. In those cases, where we have merged our shipborne dataset with the 3200 m aeromagnetic's sub-grids, it is explicitly indicated in the corresponding figure caption. The magnetic anomaly map (Figures 4 and 5) shows the typical signature associated with volcanic areas, that is, short wavelengths and high amplitudes. It is noteworthy that the greatest anomalies are not represented because they are associated with different calderas located on the islands. Although we lack onshore data, we could discern the main trends. From a detailed analysis of the magnetic map, it is possible to mark out three different magnetic areas.

The first area includes the zones close to the islands (Figure 4: inside white dashed polyline). These zones contain both the greatest amplitudes, as well as the highest frequencies. The aeromagnetic map (Socias and Mézcua, 1996) shows the real onshore situation, where every island's map is more complex than a single dipole pattern.

We could detect dipolar shape anomalies associated with every island, always showing an induced field shape, which meant that remanence was acquired during a period of normal polarity of the Earth's magnetic field (Figure 4). Younger islands (e.g., La Palma and El Hierro) appear to have the most simple characteristics, extending their magnetic influence southward.

In La Palma we mapped two large anomalies just off the island. A negative lobe projects to the north, and a positive one to the south. From the aeromagnetic map we could discern they are not

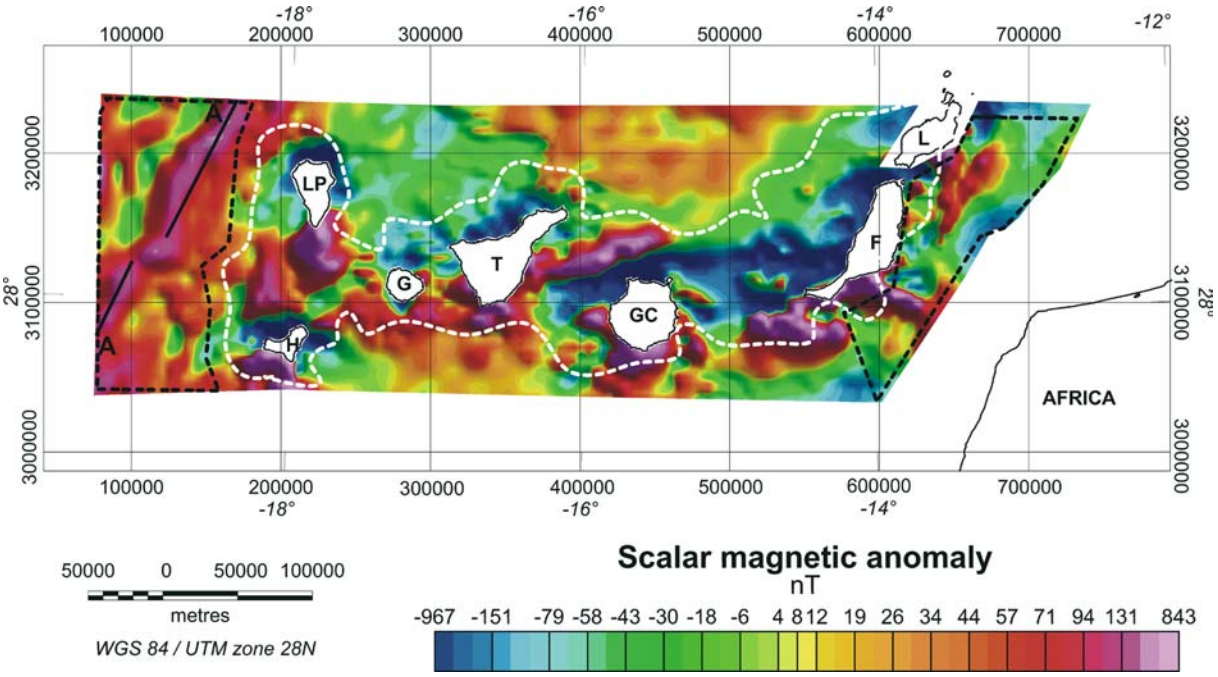

Figure 4. Detail shaded relief scalar field map of the area. Illumination from the northwest. The different magnetic areas are marked (see text for comments).

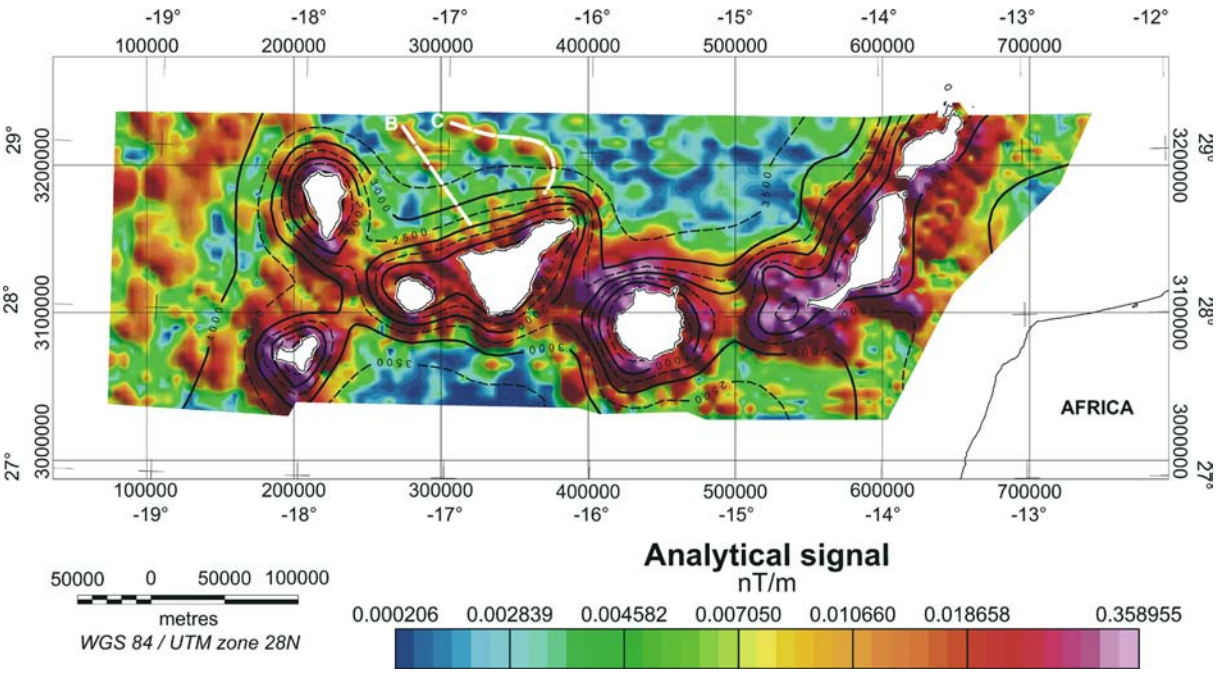

Figure 5. Shaded relief analytical signal map of the area. Illumination from the northwest. For clarification on label meaning see text. In solid and dashed lines bathymetry contour lines every 500 m.

related and belong to different magnetic sources (Instituto Geográfico Nacional, 1995).

La Gomera and Tenerife appear to be influenced by a wide, deep body and so seem to be magnetically connected. The magnetic anomalies get more complex further eastward. The central islands (Tenerife and Gran Canaria) show broader anomalies. A gently anti-clockwise trend rotation from E-W to NE-SW was detected as we moved toward the eastern island group (Fuerteventura and Lanzarote).

It is noteworthy that, between Tenerife and Gran Canaria, a very elongated anomaly is noted. Its pattern seems to be related to an inverse magnetised source. Mézcua et al. (1992) related its origin to an active fault, although this has not been verified by local seismic experiments (Funck and Schmincke, 1998).

The second area (Figure 4: inside black dashed polylines) is situated in the westernmost and easternmost part of the archipelago. The former shows strip-shaped anomalies with a nearly north or N30°E trend. This fact is more clearly demonstrated when we examine the directional derivative map (Figure 6a and b). These trends also could be detected in the eastern part of the map.

In the northern and southern central regions we recognised another magnetic style (Figure 4), and, hence, a third magnetic province (Figure 4: outside black and white dashed polyline boundaries). This third province shows great wavelength anomalies, which include more local peaks with a nearly E-W trend (Figure 6b). Particularly, in the north a great positive terrace could be discerned.

Analytical signal

To make description easier, we combined the scalar anomaly field map with the analytical signal grid. The amplitude of the analytical signal is defined as the square root of the squared sum of the vertical and two orthogonal horizontal derivatives of the magnetic field anomaly (Roest et al., 1992b; Roest and Pilkington, 1993; Salem et al., 2002).

$$|A(x,y)| = \sqrt{\left(\frac{\partial M}{\partial x}\right)^2 + \left(\frac{\partial M}{\partial y}\right)^2 + \left(\frac{\partial M}{\partial z}\right)^2}, \quad (1)$$

where $|A(x,y)|$ is the amplitude of the analytical signal, and M the magnetic anomaly field intensity. Therefore, the resulting 3D analytical signal summarises the net variation of the gradient of the magnetic anomaly field intensity in 3D.

The application of the analytic signal operator in magnetic interpretation was pioneered by Nabighan (1972). It was initially applied on profiles using a 2D approach. Lately, this method has been generalised to 3D. (Roest and Pilkington, 1993; Mohan and Babu, 1995; Hsu et al., 1998; Bastani and Pedersen, 2001). The method is a straightforward way to estimate the depth to sources or to quantify remnant effects (Salem et al., 2002).

One of the most attractive aspects of this 3D operator is the fact that its amplitude produces maxima over magnetic contacts nearly regardless of the direction of magnetisation or its induced and/or remnant character (Roest et al., 1992b; Roest and Pilkington, 1993; Keating and Sailhac, 2004). In this sense, there is no need to assume a purely induced magnetisation effect hypothesis or to discuss possible remnant vector space orientation. This simplistic approach (a purely induced magnetisation effect hypothesis) could lead to severe distortion, mainly in volcanic environments where remnant effects play an important role (Roest et al., 1992b; Roest and Pilkington, 1993; Blakely, 1995).

To analyse the area from the point of view of its magnetisation, we take advantage of the analytical signal signature (Figure 5). The analytical signal map shows three domains: One in green colour, another in yellow to orange, and the last one in red. The red region roughly corresponds to the 0.05–0.03 nT/m range in analytical signal units. The orange to yellow region is 0.02–0.01 nT/m, and the green domain delimits the smallest amplitudes ($<10^{-2}$ nT/m). If we track the bathymetry contour line map, we do not observe a single isobath fitting the green-yellow boundary but rather roughly defined zones that present higher bathymetric gradients. It is noteworthy that this operator (analytical signal) is based on first derivatives of the scalar magnetic anomaly field, highlighting boundaries according with their magnetisation contrast. These analytical signal domains present a reasonable correlation with the Free Air gravity map (Carbó et al., this issue), while this coherency is not noticeable in the Bouguer grid. All

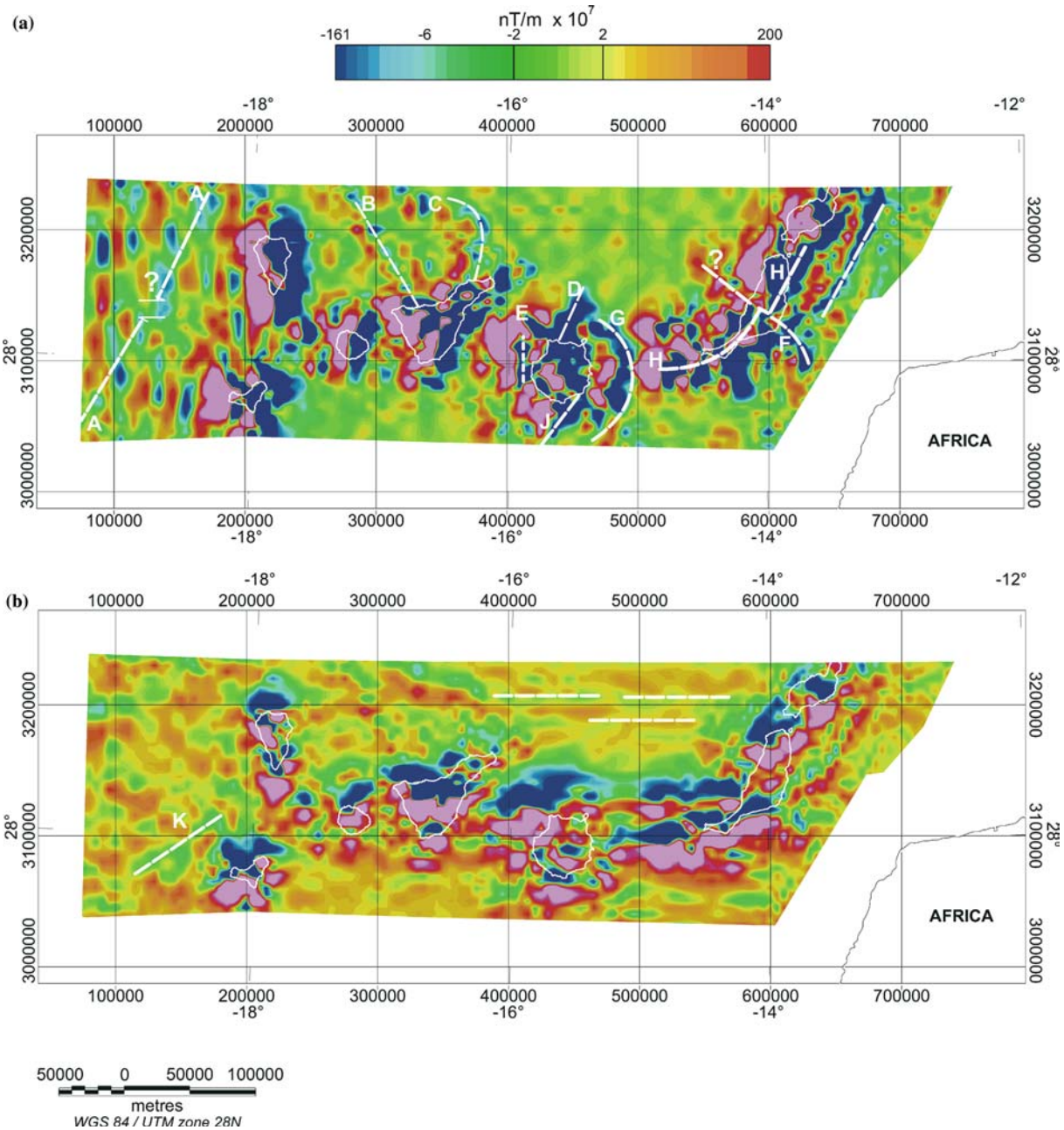

Figure 6. (a) *x*-Axis analytical signal horizontal derivative. (b) *y*-Axis analytical signal horizontal derivative. Labels A–H show suggested main magnetic alignments inferred from analytical signal horizontal directional derivatives. In order to properly track onshore features we have merged our shipborne dataset with seven 5 km × 5 km sub-grids from the 3200 m aeromagnetic grid (see section "Error estimate and data compilation" for comments).

these reasons lead us to associate the previously established analytical signal regions with topography; particularly their largest signatures are conditioned by the submarine prolongation of every island.

Depth to the source

Depth to the top: Euler Deconvolution

Magnetic methods are able to image structures at greater depths than other geophysical procedures.

In order to have a rough idea concerning the depth to the top of the different magnetic horizons, we have applied an algorithm based on the Euler Deconvolution.

The Euler Deconvolution (hereafter called "Standard Euler Deconvolution") is applied by selecting a square window of data from a total field grid and its orthogonal derivatives. Euler's equation is solved, by using the least squares method, simultaneously for each grid position within every window. The equation includes a parameter, the degree of homogeneity "N", which is usually referred to as the structural index (S.I.). This index is a measure of the rate of change with distance of a field.

This semi-automated inverse method has proved to be quite useful in several recent studies (i.e. Muszala et al., 1999; Barbosa et al., 2000). For a complete discussion on its mathematical details, refer to Thomson (1982) and Reid et al. (1990).

The "Standard Euler Deconvolution" produces a large set of solutions. Due to its vulnerability to an incorrect structural index value, and/or to the window size where it performs the inversion, solutions may be generated in areas that are free of anomalies or on the edges of anomalies. This is a difficult task to solve, especially if the area, as is the case here, is full of anomalies with quite different wavelengths. To avoid this, we have applied what we will call the "Located Euler Deconvolution".

The new method begins by calculating the analytical signal. It finds its peaks and then uses these locations for Euler Deconvolution (Blakely, 1995). This procedure produces far fewer solutions than the standard one, estimating solutions only over recognised anomalies. The final solution involves only a few depth estimates.

In the first place, in order to get an adequate S.I., we have followed the classical approach. To provide accurate solutions, the square window of data must adequately sample each anomaly on the grid. This occurs when the window size is about half the anomaly width (Durrheim and Cooper, 1998). As most of the anomalies that interest us are 30–40 km in width, a good estimate for the search window size is 20 km $\times$ 20 km. We emphasise that for our current purpose (selection of a correct S.I.) the window size is not critical.

Next, we applied an Euler Deconvolution on the grid using three different structural indices (0.5, 1.0 and 2.0). Looking for the best solution cluster (smaller scattering in the Euler Deconvolution solutions) we discarded 2.0 and 1.0 and in the end selected a value equal to 0.5. This S.I. value conditions the type of magnetic source resolved with accuracy. Because it belongs to the magnetic contact type, it has a long wavelength.

Aware that the subjectivity of the assumed "structural index" might limit the effectiveness of this technique, we have tried to compare our results with some possible depth solutions derived from other independent techniques. Ranero et al. (1997) used multichannel seismic reflection, to study three long profiles west of El Hierro and La Palma Islands. Their "B" and "C" profiles fit our needs. They detected a gradual change from smooth (eastern side of each profile) to rough basement topography (central to westernmost part of their referred profiles). Since the area we are interested in falls eastward of M21 (eastern side of their referred profiles), its basement topography is stable, and this situation favours solution quality. Top of basement is situated at 7000 ms (TWT: two-way travel time), about 5.5 km depth below sea level (b.s.l.). Urgeles et al. (1998) in the same zone, estimated, from seismic reflection, a 6900 ms (TWT) basement depth. Our "Located Euler Deconvolution" solution (based on 0.5 S.I.) estimates depth values between 4400 and 6100 m b.s.l. (Figure 7), in reasonable agreement with previously cited seismic values (Urgeles et al., 1998).

The map shows that the most significant sources lie in the 4000–6000 m range. In fact, for the great inverted dipole anomaly just NW of Gran Canaria we have obtained three depth values (b.s.l.): 5400 (label "A"), 4100 (label "B") and 3600 m (label "C"). All of them show a depth uncertainty smaller than 15% ($\pm$800 m).

Depth to bottom

The method to estimate the depth extent of magnetic sources has been treated from two points of view: by analysing the shape of isolated magnetic anomalies (e.g., Bhattacharyya and Leu, 1975), to calculate the location of the centroid, and when the depth to the top of the source is also known, then the depth to the bottom of the

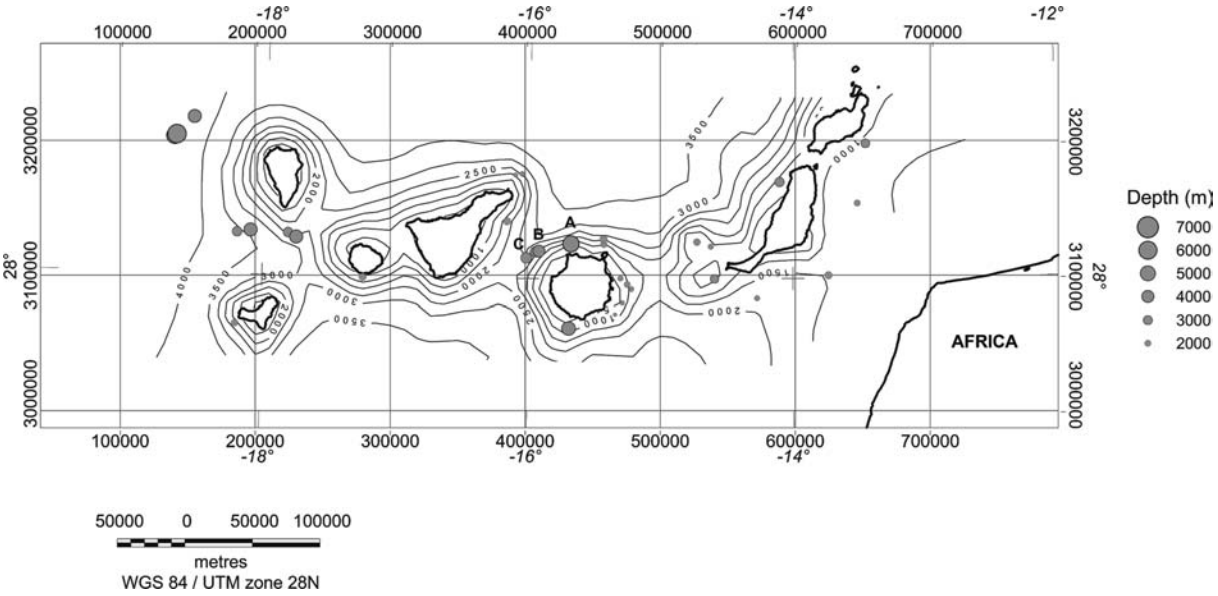

Figure 7. Euler Deconvolution solutions obtained from the scalar field map using the so-called "Located Euler Deconvolution" for SI 0.5. Bathymetry in meters. Contour interval: 500 m.

source could be obtained (Blakely, 1995). Other studies concentrated exclusively on the statistical properties of patterns of magnetic anomalies (e.g., Spector and Grant, 1970). Both methods utilise the relationship between the spectrum of the magnetic anomalies and their depths. Tanaka et al. (1999) used a similar procedure to the method of Spector and Grant (1970). The top bound and the centroid of a magnetic source were calculated from the power spectrum of magnetic anomalies and were used to estimate the basal depth of a magnetic source to magnetic anomalies from East and Southeast Asia. Shuey et al. (1977) have used the shape of radially average spectra to estimate the depth extent of magnetic sources. In particular, the position $k_{\max}$ of the maximum along the $|k|$ axis is related to the depth to the bottom of the layer according to the following equation:

$$k_{\max} = \frac{\log z_{\mathrm{b}} - \log z_{\mathrm{t}}}{z_{\mathrm{b}} - z_{\mathrm{t}}}, \tag{2}$$

where z_{b} and z_{t} are the depths (measured from the survey height) to the bottom and to the top of the layer, respectively. Although spectrum analysis provides depth estimates with simple assumptions we must highlight that these methods have fundamental limitations. Following Blakely (1995), we note that the determination of the depth to the bottom of a source (z_{b}) cannot be made without knowledge of the depth to its top (z_{t}). Second, this estimate of the depth to the bottom focus on the lowest wavenumber parts of the Fourier domain, which is quite sensitive to poorly known regional fields, and, to be able to discriminate with accuracy the peak in the spectrum, the magnetic survey must have a minimum dimension, L, which fulfils the equation:

$$L \geq \frac{4\pi(z_{\mathrm{b}} - z_{\mathrm{t}})}{\log z_{\mathrm{b}} - \log z_{\mathrm{t}}}. \tag{3}$$

Blanco-Montenegro et al. (2003) applied the method based on the $k_{\max}$ estimation on a grid centred in Gran Canaria, suggesting the presence of rocks located at mantle-like depths (from the Moho to about 23 km) which could behave as magnetic sources.

We have followed a similar procedure to that of Blanco-Montenegro et al. (2003). In order to be able to resolve the peak in the spectrum, and following Equation (3), we obtained a 347 km size for the window. We have extracted from the 3800 m upward continued grid (section "Error estimates and data compilation") three different 347 km east–west windows for three different geographical frames (Figure 8). The first one centred

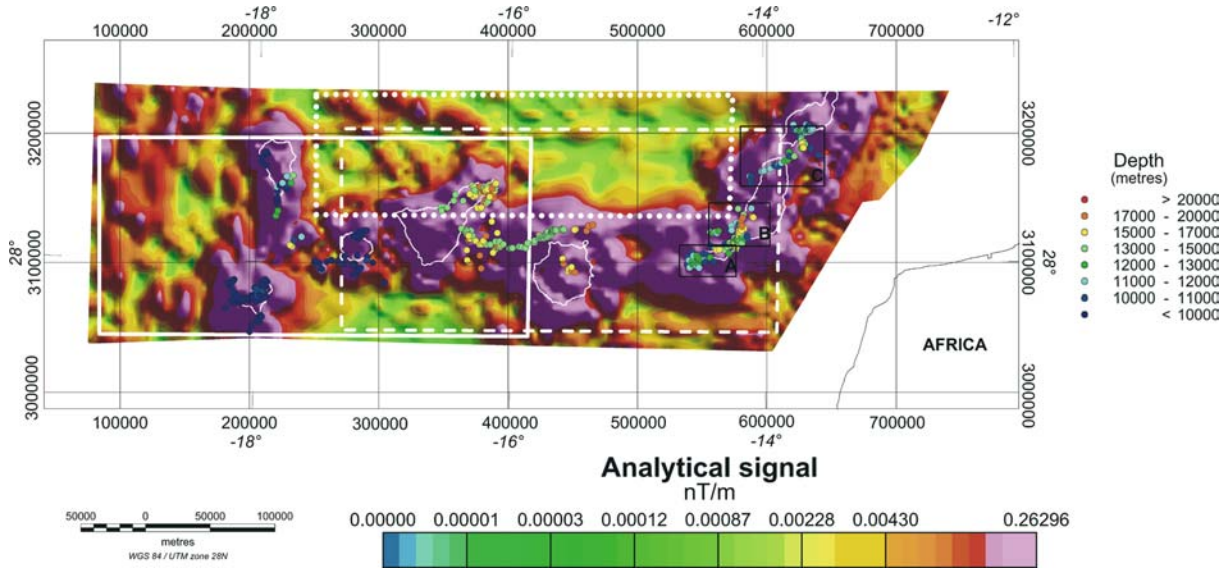

Figure 8. Shaded relief Analytical signal map of the area. Illumination from the northwest. In white (solid, dashed and spotted) lines windows used for the depth to the bottom spectral calculus (see text for comments). In black boxes, areas "A", "B" and "C" (see text for comments). Coloured circles show depth to the top of deepest magnetic sources. We have merged our shipborne dataset with seven 5×5 km sub-grids from the 3200 m aeromagnetic grid (see section "Error estimate and data compilation" for comments).

on Gran Canaria Island (dashed white line in rectangle, Figure 8), occupies the southern-central zone. The second one covers a northward parallel area (dotted white line in rectangle, Figure 8). The third one, located to the west, centred midway between El Hierro and La Palma (solid white line in rectangle, Figure 8). Their locations were not arbitrary. We have tried to isolate three different domains from the point of view of their magnetic character. One mainly concerned with volcanic rocks (Figure 8: Southern area), a second one which covers a nearly off-volcanic domain (almost oceanic) (Figure 8: Northern area), and a last one which merges both previously cited zones (Figure 8: Western area). Their location was constrained by: (a) the 347 km size area inferred from Equation (3) and (b) trying not to include areas with no data.

We have obtained the radial averaged power spectrum of every grid, which is shown in Figure 9. For the first grid (Figure 9: squares) we obtained a $k_{max} = 0.007$ cycles/km, and a $k_{max} = 0.0022$ cycles/km for the second (Figure 9: circles). The third grid (Figure 9: blue triangles) shows a different pattern. We recognised a peak, located in $k_{max} = 0.0021$ cycles/km, but a second one, with smaller amplitude, could be identified in $k_{max} = 0.0092$ cycles/km.

To obtain the z_b we have to resolve Equation (2). Following Blanco-Montenegro et al. (2003), we have assumed that $z_t = 15$ km, which is the average depth to the bottom of the oceanic crust deduced by Ye et al. (1999). The result shows that the depth to the bottom of this magnetic source (first grid) lies at 24 km b.s.l., which provides a 9 km thick magnetic source.

For the second grid we have selected a value (z_t) of 13 km, which, according to Watts et al. (1997), is the average value of the flexed crust and a typical oceanic crust domain in the area. The resolution of Equation (2) due to such a small k_{max} (0.0022 cycles/km) does not offer any numerical solution.

For the third area we lack refraction information from previous studies. We consider indirect information including: (a) The basement morphology shows no moat depression flanking the El Hierro and La Palma Islands group, and no seaward peripheral bulge is noticeable (Urgeles et al., 1998). So there is not expected to be any lithospheric flexure that could affect the Moho depth. (b) Additionally the presence of an oceanic magnetic anomaly in its westernmost margin (Figure 4) forced us to select a z_t value even smaller than 15 km. Again the resolution of Equation (2) due to such a small k_{max} (0.0021 cycles/km) does

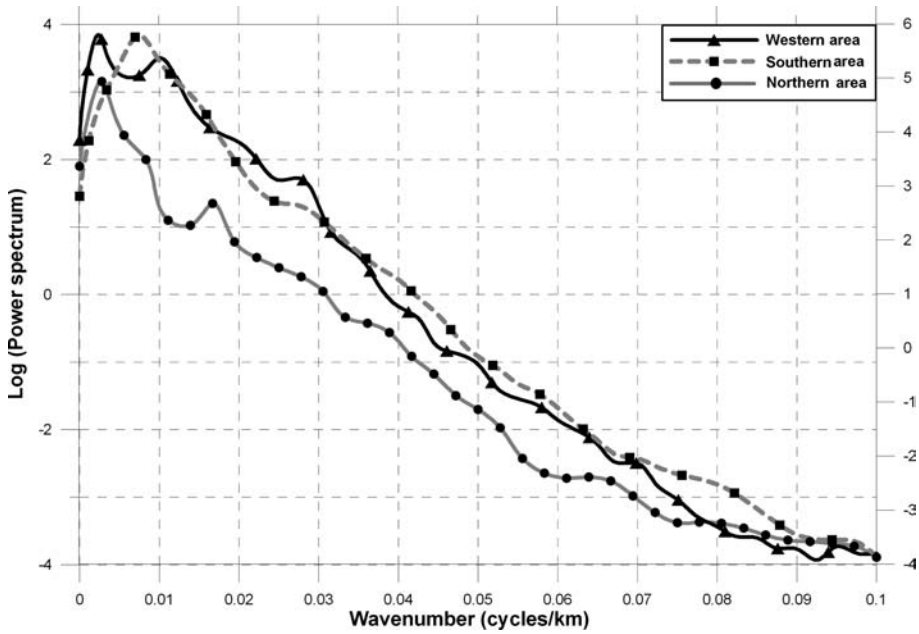

Figure 9. Radial average power spectrum from three different 347 km length windows in the east-west direction. We have merged our shipborne dataset with seven 5 km × 5 km sub-grids from the 3200 m aeromagnetic grid (see section "Error estimate and data compilation" for comments).

not offer any numerical solution. Moreover, after a careful analysis it appears to be independent of the z_t-selected value. Nevertheless as has already been mentioned, the existence of another peak at 0.0092 cycles/km needs a more specific analysis. This feature is discussed below (section "Magmatic underplating").

Alignments and trendings

We have taken advantage of the fact that the magnitude of the analytical signal is to a great extent independent of induced and remnant amplitude ratio and also of the direction of magnetisation. Accordingly, a directional derivative has been applied to two orthogonal directions, the x-axis and y-axis on the analytical signal grid, to enhance magnetic alignments (Figure 6a and b). In order to track features not only offshore but on land also, this analytical signal grid is derived from the scalar magnetic anomalies merged with seven 5 km × 5 km sub-grids from the 3200 m aeromagnetic grid, upwardly continued to 3800 m (see section "Error estimates and data compilation"). This technique highlights the main magnetic alignments around the archipelago.

Discussion of results

Lineations and magnetic trendings

The Canary Archipelago was constructed on old oceanic crust (150–170 Ma). Without accounting for the so-called "slope anomalies" (Roeser et al., 2002), its westernmost margin marks the limit of the oldest magnetic oceanic anomaly (M25) recognised in the Atlantic Ocean, as well as the beginning of the Jurassic magnetic quiet zone (Hayes and Rabinowitz, 1975; Verhoef et al., 1991; Roest et al., 1992a). Roest et al. (1992a) proposed that M25 approached the eastern flank of El Hierro and La Palma group (Figure 1).

In Figure 4, a linear trend (label A–A) appears to be associated with these western anomalies. This linear magnetic anomaly shows, to the southwest, a branch that runs sub-parallel with a nearly 21 km offset. This could be interpreted as a fracture zone (Figure 6a). We suggest, as did Verhoef et al. (1991), that this anomaly (A–A) probably corresponds to M21.

The main fractures in the islands and ocean floor have NE-SW and E-W orientations. Both cases, due to the fact they are parallel or similar

to the spreading direction of oceanic basement, are considered results of this tectonic process. These alignments could be identified in the west and north area. Likewise, the E-W trend could be discerned mainly in the north and south central areas, and always far from magnetic volcanic disturbances (Figure 6b). This could support a correlation with the tectonics of Atlantic Ocean opening.

In the eastern part of Lanzarote and Fuerteventura, NE-SW trends could also be discerned. This NE-SW pattern has suggested to different authors that the eastern islands form part of a NE-SW ridge system parallel to the African coast and related with NW-African tectonic events (Roest et al., 1992a).

Mezcua et al. (1992) noted that the eastern islands show almost a single block magnetic pattern (Figure 5). Local magnetic trends, mostly showing arc shapes, are roughly perpendicular to the inferred structural axis pointed out by Carracedo (1980), which initially runs NE-SW, turning in a clockwise direction to the E-W (Figure 6a: label "H–H").

Offshore Gran Canaria to the north and south, there are rough NE-SW magnetic alignments, and on the island's northwestern coast there is a NW-SE alignment. "D" and "E" alignments (Figure 6a) are almost perpendicular to the most remarkable anomaly of the whole survey area. Several authors have interpreted this anomaly as a fault (Anguita and Hernan, 1975; Mezcua et al., 1992; Romero et al., 2000).

The most noteworthy magnetic trend is a long wavelength feature, seen in blue (Figure 6a). This signal shows a reasonable geographical correlation with the region, which includes the strongest analytical signal amplitude (Figure 5). We interpret this wide lineation as an indicator of a region where old oceanic crust was altered by extrusive processes. This trend runs almost continuously through the archipelago and is only locally disrupted in an area between Gran Canaria and Fuerteventura (Figure 6a: label as "G"). As there is no seismic refraction information for this zone, it is difficult to assess its cause. Collier and Watts (2001) use multichannel seismic reflection profiles to determine the seismic stratigraphy of the flexural moat that flanks the Canary Islands. Their Line 15 crosses this area. Nevertheless, the top of the oceanic crust cannot be inferred due to a shallowing bathymetry and a thickening sediment layer. The geometry of the experiment (N-S) is also poor and cannot help in this regard. The Bouguer map (Carbó et al., this issue) reflects a high gradient area. Figure 5 shows a disruption in the most intense part of the analytical signal signal (red colour), but the yellow background seems to be intact. As was previously stated, it is difficult to conclude anything, although a fracture, which does not affect the whole crust, would agree with our data.

The derivative image (Figure 6a) west of El Hierro Island reveals many alignments delineating an almost north–south course, parallel to the isobaths, except the previously cited linear magnetic anomaly (Figures 4 and 6: A–A) which shows a slightly NE course. East of this oceanic magnetic anomaly we were able to observe a roughly N60°E trend (Figure 6b: label as 'K') that marks the limit of a wide area with low scalar magnetic values (Figure 4). Its inclination seems to be greater than average. Urgeles et al. (1998) recognise three seismic units west of El Hierro and La Palma bounded by strong reflectors. These seismostratigraphic units were interpreted as distal products of the volcanic evolution (erosion, mass wasting) of the archipelago. They show that the contours defined by these reflectors have a nearly N50°E trend. In contrast, the present-day seafloor contours run oblique to them, at a 30° northward angle. As was previously stated, Figure 6a shows anomaly A–A (Figure 4), which is interpreted as an oceanic anomaly. This trend shows a nearly anticlockwise 30° oblique strike with respect to the "K" alignment. Considering the strike of this alignment, and that it limits a region of low positive scalar magnetic amplitudes (to the west) and of low negative ones (to the east), we interpreted the "K" trend as a boundary between two domains: first, the transition from a marine basin, whose structure is closely related with Canary Island evolution from the point of view of its seismostratigraphic units, and, second, a typical oceanic region.

We detected a linear segment surrounding the northwestern part of Tenerife (Figure 6a: "B"), which also appears on Figure 5 (same label). These alignments join several short wavelength signals and probably are correlated with

some submarine features classified by Romero-Ruiz et al. (2000) as seamounts (their Figure 4). We suggest a probable volcanic origin for them on the basis of their high analytical signal value, in spite of the short wavelength character and a water depth greater than 3500 m. Just east, an arc-shaped set of analytical signal anomalies is well established (Figures 5 and 6: "C"). They were detected in the previously cited work and classified as seamounts (its Figure 4). We suggest a volcanic origin for these anomalies as well.

The inverse magnetised anomaly between tenerife and Gran Canaria

Funck and Schmincke (1998) analysed three seismic profiles around Gran Canaria. Their P104 and P107 profiles show a thin sediment layer on the Gran Canaria western flank (about 0.5 km thick). Considering the water depth (2500 m on average) and depth solutions b.s.l. obtained from Euler Deconvolution (section "Depth to the Top: Euler Deconvolution"): 5400 m (Figure 7: label "A"), 4100 m (Figure 7: label "B") and 3600 m (Figure 7: label "C"), we infer an upper crust location for the top of the great inverted dipole anomaly and that the causative body seems to be tilted to the NE. Nevertheless, the latter finding must be considered with caution because depth differences between the solutions are within the uncertainty associated with the measurement, roughly 1 km.

Several works (Anguita and Hernan, 1975; Mezcua et al., 1992; Romero-Ruiz et al., 2000) concluded there was an African type fault between Tenerife and Gran Canaria due to the relationship with the tectonics of the Atlas range in the African continent. This solution corresponds to a submarine fault some 50 km long with transcurrent (left-lateral) and reverse components.

Blanco-Montenegro et al. (2003) consider this linear magnetic anomaly to be an intrusive body cooled during a period of reversed polarity of the Earth's magnetic field, and suggest the possibility that it could have acted as the magmatic source of the submarine volcanism detected between Tenerife and Gran Canaria. Probably this is the most likely possibility. This magnetic source shows amplitude of 1000 nT peak to peak. After testing some forward models we estimated that an extremely thick sediment layer would be needed (greater than 6 km) so it is improbable that this magnetic anomaly could be attributed to a purely structural feature (a fault) alone. We attribute this magnetic anomaly principally to the effect of remnant magnetisation.

We consider that the fault hypothesis cannot account for this strong anomaly, at least as a major contributor because structural features usually cause small anomalies. Moreover, the Bouguer anomaly map (Carbó et al., this issue) does not show either a significant signal or a high gradient area with this strike. The possibility that the anomaly represents a reverse polarised intrusive body seems more reasonable.

Magmatic underplating

Underplating is defined as magmatic material added to the base of the Earth's crust. This geophysical feature has been reported under Hawaii (Lindwall, 1988) and the Marquesas (Caress et al., 1995) and it is still an open question and a matter of debate for the Canary Islands. Watts et al. (1997) performed a refraction seismic study in Tenerife and found no evidence of underplating beneath the island. Ye et al. (1999) performed a similar refraction seismic study, but on Gran Canaria, covering its northern and northeastern sector up to 60 km away from the coastline. A zone of magmatic underplating, as deep as 26 km, was found beneath the island, suggesting the possibility that if the underplating beneath Tenerife occurs in the same way as in Gran Canaria, that is, restricted to the main volcanic edifice without extending towards the ocean basin, the geometry of the experiment and the quality of the onshore data could have affected its detection. Dañobeitia and Canales (2000) reviewed the previously cited works (Watts et al., 1997; Ye et al., 1999) and re-examined records sections of Banda et al. (1981), reconsidering their interpretation. Dañobeitia and Canales showed a velocity depth section for both eastern islands (Fuerteventura and Lanzarote) as an alternative model. This velocity depth section shows a smooth boundary between the lower crust and a diffuse Moho, slightly deepening towards the centre of Fuerteventura. It also showed a 8–10 km thick layer in the lower crust, with a high average velocity of 7.4 km/s, overly-

ing an anomalous upper mantle velocities of 7.6–7.8 km/s. There was always a fundamental difference with other oceanic islands, like Hawaii and the Marquesas, specifically the limited radial extension of this magmatic material incorporated at the base of the crust, which does not underlie the whole volcanic edifice, but is restricted to specific parts and ends abruptly (Dañobeitia and Canales, 2000). The key question is whether magnetic data can provide some clue concerning this geophysical feature?

Two main problems appear. The first concerns the possibility that at mantle-like depths, temperatures could be low enough to remain below Curie temperature. The second concerns whether it is possible to detect such a potentially deep magnetic source. Concerning the first question, magnetisation measurements as well as petrologic studies of rock samples brought to the surface from the mantle by structural and igneous processes indicate that ferrimagnetic minerals, which are the sources of most magnetic anomalies, generally are not present in the mantle. However, the existence of sub-crustal lithospheric sources of magnetic anomalies cannot be ruled out because the possibility exists that there are localised zones where mantle processes produce ferrimagnetic minerals and that ferrimagnetic elemental iron could be present in the upper mantle (Langel and Hinze, 1998). This feature has been discussed by Blanco-Montenegro et al. (2003), taking into account the pressure dependence of the Curie point, and its relation with crustal and sediment layer thickness (Hayling and Harrison, 1986). In the case of Gran Canaria Island, this sediment layer could be 4 km thick in its northern region. Therefore, Blanco-Montenegro et al. (2003) following the Hayling and Harrison (1986) pure conductive heat flow model, concluded that a 200–300 °C temperature could be reached at the base of the crust. Considering this temperature, Blanco-Montenegro et al. (2003) stated that, if the plutonic complexes located beneath Gran Canaria were of gabbroic composition, they could contribute to the measured magnetic anomaly.

Here we summarise what is known about the Moho depth and sedimentary layer thickness knowledge for the whole archipelago before beginning an interpretation of each of the following magnetic sources analyses. The analyses seek to determine if the sources are sub-crustal, and, hence, plausible candidates for underplating. Recent local seismic studies performed in the Canary archipelago (Banda et al., 1981; Watts et al., 1997; Urgeles et al., 1998; Ye et al., 1999) suggest that a value of 15 km for the Moho depth, and a 5 km sedimentary cover thickness could be considered representative for the central islands (Gran Canaria and Tenerife). A different situation occurs in the eastern islands, where seismic models (Banda et al., 1981; Dañobeitia and Canales, 2000) show a smooth boundary between the lower crust and a diffuse Moho, which deepen slightly towards the centre of Fuerteventura, reaching a maximum depth of 24–25 km. The upper limit for the sediment cover thickness is 5 km.

With respect to the western islands (La Palma and El Hierro), we note that the sedimentary layer on their western submarine flanks reaches nearly 5 km thickness (Ranero et al., 1997; Urgeles et al., 1998). As there is no refraction and reflection seismic data for the islands' immediate surroundings, nothing can be said about the Moho and the actual thickness of the sedimentary layer, except that it seems implausible that the Moho is deeper than 15 km. The reason, according to the Bouguer anomaly map (Carbó et al., this issue), is that the islands are located on the westernmost margin of a transitional crust, and from 342° meridian westward, isolines delineate a strongly north-southward component, showing Bouguer anomalies ranging in accord with oceanic values.

This situation leads us to believe that in the offshore marine area an underplated source in Canary Archipelago could have a magnetic signature and thus appear on the magnetic anomaly map. In the same way, we could expect there to be a similar source attached to a Fuerteventura and Lanzarote islands, possibly at a depth of 13–15 km, with a magnetic signature. However, its sub-crustal origin cannot be asserted "a priori".

Determining whether or not such a potentially deep magnetic source can be detected is key to answering the second question. From the depth to the bottom analysis (section "Depth to bottom") we have obtained three profiles that represent the radial average power spectrum of three different grids (Figure 9). In the southern area, the peak situated at $k_{max} = 0.007$ cycles/km could

be compatible with the presence of a magnetic source with a depth to the bottom of 24 km b.s.l., indicating a nearly 9-km-thick magnetic body. As was stated in section "Depth to Bottom", the northern area spectrum is not compatible with the presence of any sub-crustal source. Since in the Canary Archipelago any magmatic underplating evidence has always been connected with volcanic islands (Ye et al., 1999; Dañobeitia and Canales, 2000), this result is supported by the absence of any island in its interior.

The third grid, in the westernmost area, shows two peaks, one at $k_{max} = 0.0021$ cycles/km, and a second one at $k_{max} = 0.0092$ cycles/km. Although the first is the greatest, in power terms, the latter shows a slightly lower value: 1.39 times in power terms units and must not be discarded. We interpret this anomalous behaviour to the fact that this mathematical procedure is based on a probabilistic behaviour of the distribution of magnetisation, which assumes magnetisation to be isotropic, that is, its statistical behaviour is the same in all horizontal directions (x and y) (Blakely, 1995). From Figure 8, we see that this is not correct for the westernmost window, which shows an unbalanced situation with a lineated anomaly in its western border, and strong magnetic anomalies in its right half associated with volcanic islands (Tenerife, Gomera, Hierro and La Palma). In this sense, we must take into account the presence of the other peak ($k_{max} = 0.0092$ cycles/km). If we solve for a $z_t = 15$ km, Equation (2) provides a depth to the bottom solution at 20 km b.s.l. This value corresponds to a 5 km thick magnetic source, but it is surely influenced by the variability of the Moho depth in this area, where we link a transitional crust domain and an oceanic one. If we solve with $z_t = 13$ km, which could better fit an average Moho depth value in the westernmost frame, the solution turns out to be 22.5 km, which corresponds with an almost 10 km thick magnetic source and shows reasonable agreement with seismic values (Ye et al., 1999).

Another interesting aspect is the location and distribution of all these possible sub-crustal magnetic sources. We have taken advantage of the Euler Deconvolution algorithm, but must take into account some drawbacks: (a) This algorithm incorporates three gradients. This fact makes it sensitive to noise, which could affect the quality of the solution by introducing fluctuations and incoherencies into solutions between adjacent grids. (b) We are interested in a potential deep source that could have a low magnetic signature that must be isolated. (c) Previous geophysical studies of the archipelago have shown the local character of the signal we are seeking. It seems to be associated always with islands (Tenerife, Gran Canaria, and possibly Lanzarote and Fuerteventura), and shows a nearly 10 km vertical extent.

To overcome all these drawbacks we have applied the "Standard Euler Deconvolution" instead of the so-called automatic "Located Euler Deconvolution" procedure. As was stated for drawback (b), the signal we are interested in does not necessarily show a high magnetisation contrast, and this semi-automated method, based on the detection of analytical signal peaks, could be inadequate.

Drawback (c), states that the underplated body is limited in both the vertical and horizontal planes. We have selected a high SI. An index equal to 3 should represent an ideal body, a sphere. An index equal to 2 can describe a body with only one infinite dimension. We believe this is an intermediate case between both and is best seen in an index of 2.5 (Reid et al., 1990).

Another significant question concerns window size. This method employs overlapping windows, resolving Euler's Equation for each. The window of data must adequately sample each anomaly on the grid. This window must be large enough to include the entire anomaly being analysed, but not so large that it could contain multiple anomalies, which could lead to confusing solutions. Additionally, a small window could be more sensitive to noise as the number of data points in the window is decreased.

As the anomaly associated with every island has different sizes, we have selected a variable window size, different for every island. To set a criteria to determine the window size we have used the analytical signal wavelength. We have selected, as a balance, a window size equal to half the characteristic analytical signal wavelength associated with every island. This window size goes from a 15 km $\times$ 15 km value for Gomera island to 40 km $\times$ 40 km for Gran Canaria and Tenerife islands. With all these conditions, especially the window size criteria, we

have highlighted the deepest sources. To reduce topographic influences, as well as to highlight deeper sources, we have upwardly continued the mixed "shipborne plus aeromagnetic" 3800 m grid (see section "Error estimate and data compilation" for details) to 5000 m. Those solutions that show a 15% depth uncertainty or that fall outside their correspondent window cell are discarded.

Gran Canaria: This island shows an homogeneous set of solutions, all of them deeper than 15 km. It is observed that they appear to be located in the northern half of the island (central, NNW and NNE sector). This result shows good agreement with constraints from seismic methods (Ye et al., 1999).

Tenerife: A similar pattern as that found at Gran Canaria was observed. A clear cluster of solutions is displayed in the NE sector of the island (Anaga Peninsula). The data set of solutions obtained on the eastern offshore side of the island shows a certain spatial variability. Nevertheless, most of solutions (72%) fall into the 14–16 km segment, which takes into account the uncertainty associated with this type of depth estimation and fits quite well with the Moho under this island.

No refraction seismic experiments have been performed in this area. Additionally, in roughly the same area, the Bouguer map (Carbó et al., this issue) presents a wide terrace whose boundary is set by the 200 mGal isoline. This area contains a body with a lower than expected density, considering the monotonic transition from a continental domain to an oceanic one observed to have a strong north-south component. This transition basically shows a change in Moho depth, apparently disrupted in this area due to the presence of the slightly low-density body that magnetic solution indicates could have its top at Moho-like depths. We suggest that it is plausible to consider the body to be magmatic underplating.

We observed a set of solutions, marking depths from 13.5 to 14.5 km, that link the island of Gran Canaria and Tenerife. This set of solutions shows a remarkably linear trend and a reasonable correlation with the inverse magnetic anomaly track (Figure 4). Ye et al. (1999) inferred that both ocean islands' uplift and evolved volcanism were caused by the same process of magmatic underplating. The up-welling magma did not make it to the surface but, due to its density and rheology, was trapped below the crust and fractionated into more dense, basic magma added to the base of the crust and lighter, more evolved magma able to continue rising. This explanation agrees with the interpretation of the inverse magnetised body between Tenerife and Gran Canaria. In the section "The inverse magnetised anomaly between Tenerife and Gran Canaria" we conclude that the possibility that this anomaly represents a reverse polarised intrusive body seems more reasonable. With an "Extended Euler Deconvolution", we located its top at an upper crust level. We suggest that the linear magnetic anomaly we reflect in the total field anomaly grid could be a consequence of two contributions, one deeper than the other: one with a top at upper crust like depths, and the other at a sub-crustal level, but both linked in their origin. This possibility offers another clue that cannot be discarded as long as we do not have refraction seismic information for this particular area. There is the possibility that the same plutonic complex links both Tenerife and Gran Canaria.

La Gomera: The entire set of depth solutions falls below 10 km, in fact 88% are less than 7.6 km. There is lack of information concerning Moho depth at La Gomera, nevertheless a 15 km average value for crustal thickness at Tenerife seems to support a crustal origin (possibly upper crust) for causative magnetic bodies.

El Hierro: A similar analysis could be applied on El Hierro. The depth to the top of the causative source ranges from a minimum 6.5 km to 11.2 km depth. Most of them (66%) are grouped around 9 km b.s.l. Its westernmost position makes it difficult to conclude anything about its crustal or upper mantle origin.

La Palma: There are some depths values situated on the NW side of the island. They are concentrated around 6.1 km depth (minimum = 5.1 km; maximum: 7.4 km). On its southern side we found certain variability, in fact there was a 10.2 km average depth value, and 13.1 km and 8.2 km as the maximum and minimum depths to the top of the magnetic source, respectively. We inferred a step of nearly 4 km between north and south solutions. We suggest a crustal origin for the north set of depths, but the south

group of solutions deserves a more careful analysis. In spite of the depth variability, we see that solutions are clearly deeper than in the north (almost 4 km). Once again we lack seismic information to locate the Moho. Offshore La Palma we could see another depth solution cluster. They show similar statistics with regard to the depth solution group just mentioned. Moreover, in the scalar field map (see Figure 4) there is a southward positive anomaly. In the same sense, the Bouguer map (Carbó et al., this issue) shows a non-absolute minimum anomaly in the area. Taking both facts into consideration, the possibility of an anomalous mantle seems plausible and coherent.

Fuerteventura and Lanzarote: The depth solution shows variability. This has to do with the estimative nature of the algorithm and that is why we prefer to use statistical terms. In order to simplify the discussion we have drawn three boxes (Figure 8, labelled A, B and C). These groups of boxes have three different average "depth to the top of the deepest magnetic sources" levels. In box "A", from a set of solution of 26 values, we have obtained 11.5 km as a mean depth value, with maximum and minimum depths equal to 12.8 and 10.1 km, respectively. In box "B", from a set of solution of 41 values, we have obtained 13.2 km as the mean depth value, with maximum and minimum depths equal to 16.0 and 10.0 km, respectively. It is worth pointing out that only six values were greater than 11 km. In box "C", from a set of solution of 55 values, we have obtained 11.5 km as a mean depth value, maximum and minimum depths equal to 15.5 and 10.1 km, respectively. Again, it is important to stress that only six values are greater than 11 km.

We could discern a nearly 2 km jump between "A" and "B" frames, and in "C" a similar horizon to "A". Dañobeitia and Canales (2000) reexamined old seismic records sections from Banda et al. (1981) suggesting the presence of underplated material but denote, however, a slight deepening of Moho boundary, as well as for the upper to lower crust limit, towards the centre of Fuerteventura as a consequence of a flexure. Our results agree are consistent with that conclusion.

Figure 8 shows an absence of solutions from the south to the north set. Our depth solutions are located over the higher analytical signal amplitude sources. We consider that these places mark the locations where upwelling magmas interact with an old oceanic crust, disrupting and magnetising it and that this process does not affect the whole volcanic edifice but is instead restricted to specific parts. Additionally, in Figure 6a we observed the presence of a high gradient in the NW-SE direction in this zone (Figure 6a: label "F"). It may be an indicator of a Fault. This aspect could justify the absence of solutions in this area, and the existence of a nearly 1.7 km vertical average offset between "B" and "C" frame solutions.

Conclusions

In this paper we have described the main geomagnetic characteristics of ZEEE in the region of the Canary Islands. In the first part, we concentrated mainly on purely technical aspects including: data acquisition and treatment, instrumentation, and quality of the surveys. The analysis of this data set has led to several conclusions, summarised as follows.

From a spectral analysis we have obtained evidence, which shows the possible existence of magnetic sources. The evidence locates the depth to the bottom at a sub-crustal level. They seem to be located especially in the southern islands area (Figure 8). They show a 24 km b.s.l depth extent, which indicates a 9-km-thick magnetic source.

Additionally, we applied a Located Euler Deconvolution algorithm to the scalar magnetic anomaly grid and obtained three depth values b.s.l. 5400 m (label "A"), 4100 m (label "B"), and 3600 m (label "C"), for the inverse magnetised anomaly between Gran Canaria and Tenerife. These results suggest a top upper crust location for the basement top of the magnetic body.

Using a Standard Euler Deconvolution algorithm we have located the deepest magnetic sources in the archipelago, analysed their spatial distribution and their possible sub-crustal magnetic origin. We have found evidence of these sources, particularly at Gran Canaria and Tenerife. Their distribution seems to be restricted to specific areas, in good agreement with previous seismic studies (Ye et al., 1999; Dañobeitia and

Canales, 2000). Fuerteventura and La Palma show a cluster of solutions at the 11.5–13.2 km range, and 10.2 km b.s.l. depth, respectively. Seismic studies (Dañobeitia and Canales, 2000) show that at Fuerteventura there is a diffuse Moho boundary. At La Palma there is no refraction seismic data to delineate the Moho. In this situation, nothing can be inferred yet, concerning the sub-crustal origin of these deep magnetic sources.

The same analysis (Euler Deconvolution) provided evidence which suggests the possibility that the inverse magnetised body between Tenerife and Gran Canaria could tie together two contributions: one at upper crust levels and the other, deeper, seems to be mantle in its origin. The continuity of the solution data set suggests another possibility, that is, that Tenerife and Gran Canaria could be connected at depth by the same plutonic complex. Concerning the westernmost islands we have found no evidence that supports the presence of magmatic underplating under either La Gomera or El Hierro islands.

Additionally, south of La Palma a clear cluster of solutions appears. Bouguer map (Carbó et al., this issue) reflects non-absolute minimum values at the area. These factors support the existence of an anomalous mantle, which starts in the south of La Palma and extends nearly 27 km southeastwardly.

The Euler Deconvolution analysis shows a gap in the depth solutions at Fuerteventura and Lanzarote. The magnetic alignment map (Figure 6a) shows a high gradient in the NW–SE direction, in the gap area. We consider this to be an indication of a major fracture.

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